(c, p) **slope law: observability matrix**

1 Model

Gain law (slope only)

$$\bar{w} = \frac{w}{R}, \qquad \bar{w}_s = \text{visible surface}, \qquad h = \bar{w} - \bar{w}_s, \qquad \hat{\bar{w}}[k] = \bar{w}_d[k] - \delta\hat{\bar{w}}_3[k], \qquad \hat{h}[k] = \hat{\bar{w}}[k] - \bar{w}_s$$

$$\bar{a}'(\hat{h};\, c, p) = \frac{1-c}{\hat{h}^p}, \qquad \frac{\partial \bar{a}'}{\partial c} = -\frac{1}{\hat{h}^p}, \qquad \frac{\partial \bar{a}'}{\partial p} = -\,\bar{a}'\,\ln\hat{h}, \qquad \frac{\partial \bar{a}'}{\partial \hat{h}} = -\frac{p\,\bar{a}'}{\hat{h}}$$

State equation (M = the step the estimator believes the particle makes)

$$\begin{aligned}
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k] \\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c \cdot \delta\bar{w}_3[k] - F_{dw}[k] \cdot e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\bar{a}_w[k+1] &= \bar{a}_w[k] + \bar{a}'\big(\hat{h}[k];\, c, p\big)\, M[k] \\
c[k+1] &= c[k] \\
p[k+1] &= p[k]
\end{aligned}$$

$$M[k] = \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k], \qquad F_{dw}[k] \approx \big(1 + (1-\lambda_c)d\big)\,\frac{\Delta\bar{w}_d[k]}{\hat{\bar{a}}_w[k]} \qquad \text{(constant velocity)}$$

Output equation ($\nabla_d\bar{w}_d[k] = \bar{w}_d[k] - \bar{w}_d[k-d]$)

$$\begin{aligned}
y_1[k] &= \delta\bar{w}_1[k] + n_w[k], \qquad R_1 = \mathrm{Var}(n_w) \\
y_2[k] &= \bar{a}_w[k-d] + n_a[k], \quad R_2 = \mathrm{Var}(n_a), \qquad \hat{y}_2[k] = \hat{\bar{a}}_w[k] - \bar{a}'\big(\hat{h}[k];\, \hat{c}[k], \hat{p}[k]\big)\,\nabla_d\bar{w}_d[k]
\end{aligned}$$

Estimation error and output error ($e_\theta := \theta - \hat{\theta}$)

$$\mathbf{e}[k] = \big[\, e_{\delta\bar{w}_1},\ e_{\delta\bar{w}_2},\ e_{\delta\bar{w}_3},\ e_{\bar{a}_w},\ e_c,\ e_p \,\big]^\top, \qquad e_{y_1}[k] = y_1[k] - \delta\hat{\bar{w}}_1[k], \qquad e_{y_2}[k] = y_2[k] - \hat{y}_2[k]$$

Error propagation (no noise, coefficients at the estimate)

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k], \qquad \mathbf{e}[k_0+j] = \mathbf{\Phi}(k_0+j, k_0)\,\mathbf{e}[k_0], \quad \mathbf{\Phi}(k_0+j, k_0) = \mathbf{F}_e[k_0+j-1]\cdots\mathbf{F}_e[k_0]$$

$$\mathbf{F}_e[k] = \begin{bmatrix}
0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & \lambda_c & -F_{dw}[k] & 0 & 0 \\
0 & 0 & \bar{a}'(1-\lambda_c) + \frac{p\,\bar{a}'M[k]}{\hat{h}[k]} & 1 + \bar{a}'F_{dw}[k] & -\frac{M[k]}{\hat{h}[k]^p} & -\bar{a}'\,\ln\hat{h}[k]\; M[k] \\
0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1
\end{bmatrix}$$

$$\mathbf{H}[k] = \begin{bmatrix}
1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{p\,\bar{a}'\,\nabla_d\bar{w}_d[k]}{\hat{h}[k]} & 1 & \frac{\nabla_d\bar{w}_d[k]}{\hat{h}[k]^p} & \bar{a}'\,\ln\hat{h}[k]\,\nabla_d\bar{w}_d[k]
\end{bmatrix}$$

2 Observability matrix

Leading order (as in the c-only document)

$$\beta_k^c := \frac{M[k]}{\hat{h}[k]^p}, \qquad \beta_k^p := \bar{a}'\,\ln\hat{h}[k]\;M[k] = (1-c)\ln\hat{h}[k]\;\beta_k^c$$

$$\gamma_k^c := \frac{\nabla_d \bar{w}_d[k]}{\hat{h}[k]^p}, \qquad \gamma_k^p := \bar{a}'\,\ln\hat{h}[k]\;\nabla_d\bar{w}_d[k]$$

$$\mathbf{F}_e[k] \to \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw}[k] & 0 & 0 \\ 0 & 0 & 0 & 1 & -\beta_k^c & -\beta_k^p \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{H}[k] \to \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & \gamma_k^c & \gamma_k^p \end{bmatrix}$$

Errors at step k_0, output errors j steps later

$$\begin{bmatrix} e_{y_1}[k_0+j] \\ e_{y_2}[k_0+j] \end{bmatrix} = \mathbf{H}[k_0{+}j]\;\mathbf{\Phi}(k_0{+}j, k_0)\,\mathbf{e}[k_0], \qquad s_j^c := -\sum_{i=0}^{j-1}\beta_{k_0+i}^c + \gamma_{k_0+j}^c, \qquad s_j^p := -\sum_{i=0}^{j-1}\beta_{k_0+i}^p + \gamma_{k_0+j}^p$$

The matrix ($F_i := F_{dw}[k_0+i]$)

$$\begin{bmatrix} e_{y_1}[k_0] \\ e_{y_2}[k_0] \\ e_{y_1}[k_0+1] \\ e_{y_2}[k_0+1] \\ e_{y_1}[k_0+2] \\ e_{y_2}[k_0+2] \\ e_{y_1}[k_0+3] \\ e_{y_2}[k_0+3] \\ e_{y_1}[k_0+4] \\ e_{y_2}[k_0+4] \\ \vdots \\ e_{y_1}[k_0+j] \\ e_{y_2}[k_0+j] \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & s_0^c & s_0^p \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & s_1^c & s_1^p \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & s_2^c & s_2^p \\ 0 & 0 & \lambda_c & -F_0 & 0 & 0 \\ 0 & 0 & 0 & 1 & s_3^c & s_3^p \\ 0 & 0 & \lambda_c^2 & -\lambda_c F_0 - F_1 & F_1\beta_0^c & F_1\beta_0^p \\ 0 & 0 & 0 & 1 & s_4^c & s_4^p \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \lambda_c^{j-2} & -\sum_{i=0}^{j-3}\lambda_c^{j-3-i}F_i & \sum_{i=0}^{j-3}\lambda_c^{j-3-i}F_i\sum_{l<i}\beta_l^c & \sum_{i=0}^{j-3}\lambda_c^{j-3-i}F_i\sum_{l<i}\beta_l^p \\ 0 & 0 & 0 & 1 & s_j^c & s_j^p \end{bmatrix}}_{\mathbf{O}(k_0)} \begin{bmatrix} e_{\delta\bar{w}_1}[k_0] \\ e_{\delta\bar{w}_2}[k_0] \\ e_{\delta\bar{w}_3}[k_0] \\ e_{\bar{a}_w}[k_0] \\ e_c[k_0] \\ e_p[k_0] \end{bmatrix}$$

The six columns (F_{dw}, β, γ constant)

	y_1 rows ($j = 0, 1, 2, \dots$)	y_2 rows ($j = 0, 1, 2, \dots$)
$e_{\delta\bar{w}_1}$	$1,\ 0,\ 0,\ 0,\ \dots$	$0,\ 0,\ 0,\ \dots$
$e_{\delta\bar{w}_2}$	$0,\ 1,\ 0,\ 0,\ \dots$	$0,\ 0,\ 0,\ \dots$
$e_{\delta\bar{w}_3}$	$0,\ 0,\ 1,\ \lambda_c,\ \lambda_c^2,\ \dots \to 0$	$0,\ 0,\ 0,\ \dots$
$e_{\bar{a}_w}$	$0,\ 0,\ 0,\ -F_{dw},\ -F_{dw}(1+\lambda_c),\ \dots \to -\dfrac{F_{dw}}{1-\lambda_c}$	$1,\ 1,\ 1,\ \dots$
e_c	$0,\ 0,\ 0,\ 0,\ F_{dw}\beta^c,\ F_{dw}\beta^c(2+\lambda_c),\ \dots$	$\gamma^c,\ -\beta^c+\gamma^c,\ -2\beta^c+\gamma^c,\ \dots$
e_p	$0,\ 0,\ 0,\ 0,\ F_{dw}\beta^p,\ F_{dw}\beta^p(2+\lambda_c),\ \dots$	$\gamma^p,\ -\beta^p+\gamma^p,\ -2\beta^p+\gamma^p,\ \dots$

3 What the matrix shows

$e_{\bar{a}_w}$, e_c, e_p in the y_2 rows

$e_{\bar{a}_w}:\ 1,\ 1,\ 1,\ \ldots \qquad e_c:\ s_0^c,\ s_1^c,\ \ldots \qquad e_p:\ s_0^p,\ s_1^p,\ \ldots$

$$\det\begin{bmatrix}1 & s_j^c\\ 1 & s_{j+1}^c\end{bmatrix} = -\beta_j^c, \qquad \det\begin{bmatrix}1 & s_j^p\\ 1 & s_{j+1}^p\end{bmatrix} = -\beta_j^p$$

$$\det\begin{bmatrix}s_j^c & s_j^p\\ s_{j+1}^c & s_{j+1}^p\end{bmatrix} = \beta_j^c\, s_j^p - \beta_j^p\, s_j^c = \beta_j^c\Big(s_j^p - (1-c)\ln\hat{h}_j\ s_j^c\Big)$$

$$\det\begin{bmatrix}1 & s_j^c & s_j^p\\ 1 & s_{j+1}^c & s_{j+1}^p\\ 1 & s_{j+2}^c & s_{j+2}^p\end{bmatrix} = \beta_j^c\,\beta_{j+1}^p - \beta_j^p\,\beta_{j+1}^c = (1-c)\,\beta_j^c\,\beta_{j+1}^c\left(\ln\hat{h}_{j+1} - \ln\hat{h}_j\right)$$

Hold

$$\Delta\bar{w}_d = 0: \qquad \gamma^c = \gamma^p = 0, \qquad M = (1-\lambda_c)\,\delta\hat{\bar{w}}_3, \qquad s_j^c \approx 0,\ \ s_j^p \approx 0$$

Constant velocity, $\Delta\bar{w}_d = (h_0 - h_b)\,T_s/T_d$; coefficients at the plane init $\hat{c} = 0.05$, $\hat{p} = 2$

$$s_j^c \approx \frac{1}{\hat{h}_j} - \frac{1}{\hat{h}_0}, \qquad s_j^p \approx (1-c)\left[\frac{1+\ln\hat{h}_j}{\hat{h}_j} - \frac{1+\ln\hat{h}_0}{\hat{h}_0}\right], \qquad r_j := \frac{s_j^p}{s_j^c}$$

$c = 0.05,\ \ \hat{h}_0 = 22.2:$

$\hat{h}_j$	10	3	1.1
s_j^c	0.055	0.29	0.86
s_j^p	0.14	0.49	0.77
r_j	2.5	1.7	0.90

$$e_{y_2}[k_0 + j] = e_{\bar{a}_w} + s_j^c\big(e_c + r_j\, e_p\big)$$

weak direction in $(e_c, e_p):\ \propto (1,\ -1/r_j) \approx (0.7,\ -0.7)$ where s_j^c is large

Plane init on the cell wall

$$(\Delta c, \Delta p) = (+0.64,\ -0.53), \qquad |(\Delta c, \Delta p)| = 0.83$$

seen by $y_2:\ \ \Delta c + 0.9\,\Delta p = 0.16, \qquad$ along $(1, -1/0.9):\ \ 0.82$